INSURA

AI-Based Gig Worker Insurance System

Project Report

Team Details

Team Members:
1. Madineedi Jaya Sai Nikhilendra
2. Vundavalli Iswarya
3. Addagarla Ramya Krishna
4. Javvadi Venkata Naga Sudheer
5. Ganugula Vishnu

College: **Koneru Lakshmaiah Education Foundation**

Table of content

Team Details 2
Introduction 4
Problem Statement 5
Persona 6
Solution 7
Technologies Used: 8
Pricing Model 9
Adversarial Defense & Anti-Spoofing Strategy 13
Workflow 15
Conclusion 16
Presentation video 17

Introduction

India's gig economy has grown rapidly, driven by millions of delivery partners working with platforms like Zomato, Swiggy, Amazon, and others. These workers play a critical role in supporting urban lifestyles and digital commerce. However, their earnings remain highly vulnerable to external disruptions such as extreme weather conditions, pollution, and natural disasters, which can significantly reduce their working hours and income.

Despite their importance, gig workers currently lack adequate financial protection against such unpredictable events. This creates a pressing need for innovative insurance solutions tailored to their unique work patterns and risks.

To address this gap, this project proposes an **AI-powered parametric insurance platform** designed specifically for gig workers in India. By leveraging real-time data and intelligent automation, the platform aims to provide quick, transparent, and reliable income protection through automated payouts, while ensuring simplicity, affordability, and fraud-resistant operations aligned with a weekly earnings model.

Problem Statement

Gig workers working for platforms like Zomato, Swiggy, and Blinkit depend on daily orders for income.

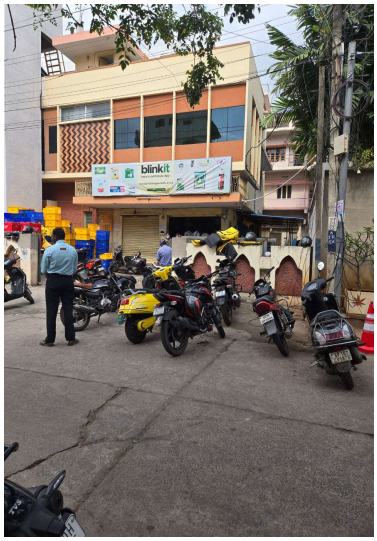
However, during:

- Heavy rain
- Floods
- Heatwaves
- Curfews
- Government shutdowns (Bundh)

they cannot work and lose their daily income.

Most gig workers do not have insurance or income protection, leaving them financially vulnerable.

Persona



During our field interaction, we engaged with several gig workers to better understand the challenges they face in their daily work. From these conversations, we developed a representative persona that reflects the common experiences shared by many of them.

The gig workers explained that their income is entirely dependent on the number of tasks or deliveries they complete each day, with no fixed or guaranteed salary. This makes their earnings highly unpredictable and unstable. External factors such as heavy rain, extreme heat, pollution, traffic congestion, and unforeseen disruptions significantly affect their ability to work, thereby directly impacting their daily income.

They also highlighted that even a short period of disruption, such as a few days without work, can lead to considerable financial strain, making it difficult to manage basic living expenses. Despite working long and irregular hours, gig workers lack any form of financial safety net or compensation during such interruptions.

These insights helped us understand the extent of financial instability faced by gig workers and emphasized the urgent need for a reliable income protection system to support them during periods of disruption.

Solution

To solve this problem, we propose a **digital platform that provides weekly income protection using a parametric insurance model**. This platform allows gig workers to subscribe to affordable weekly plans and receive automatic payouts when predefined conditions are met.

The system works in a simple and user-friendly manner. First, workers register on the platform using their phone number and link their gig service account (like Swiggy or Zomato). Then, they choose a weekly subscription plan—Basic, Standard, or Pro—based on their needs.

The platform continuously monitors real-time data such as weather conditions, government alerts, and city shutdowns using APIs. If certain thresholds are crossed—such as rainfall above 120mm, extreme heat, curfew announcements, or flood warnings—the system automatically triggers compensation payouts. This removes the need for manual claims and ensures quick and transparent financial support.

AI and Machine Learning (Usage):

1. Machine Learning for Predictive Risk Scoring

2. Time-Series Forecasting for Dynamic Weekly Premiums

3. Natural Language Processing (NLP) for Social Disruption Detection

4. Anomaly Detection for Fraud Prevention

The system uses **AI and Machine Learning** to enhance its performance. AI helps in predicting climate risks, detecting fraudulent activities, and dynamically adjusting premium rates based on location, historical weather data, and user behavior. This makes the platform more intelligent, efficient, and fair.

Technologies Used:

Technologically, the platform is built as a **web application** using React and CSS for the frontend, Spring Boot for the backend, and MySQL for data storage. It is deployed on AWS using services like Lambda and API Gateway, and integrates external APIs such as OpenWeather for real-time data.

Pricing Model

Dynamic Pricing & Probability Model for Insura

1. Disruption Factors

Environmental Factors

- Rainfall
- Extreme temperature
- Flood alerts
- Pollution

Social Factors

- Curfews
- Strikes
- Zone restrictions

2. Probability Calculation

Step 1: Data → Risk Scores (0–1)

Rainfall (mm/hr)	Score
0–5	0.1
5–20	0.3
20–50	0.5
50+	0.7

Step 2: Environmental Risk

Environmental Risk = Average of all environmental scores

Step 3: Social Risk

Social Risk = Average of all social scores

Step 4: Final Probability

Total Probability = (0.7 × Environmental Risk) + (0.3 × Social Risk)

3. Parametric Trigger Conditions

Payout is automatically triggered when:

- Rainfall exceeds threshold
- Temperature exceeds threshold
- Flood alert is issued
- Government restrictions are applied

4. Pricing Model

Risk-Based Plans

Risk Level	Premium (Weekly)	Daily Payout	Max Weekly
Low	₹125	₹275	₹700
Medium	₹150	₹325	₹700
High	₹175	₹325	₹700

Optional Fixed Plans

Plan	Premium	Daily Payout	Max Weekly
Basic	₹100	₹200	₹400
Standard	₹200	₹400	₹800

Usage Logic

- System assigns a default risk-based plan (Low / Medium / High)
- Users can optionally select a fixed plan
- Fixed plans ignore area-based pricing but still follow disruption triggers

5. Pricing Logic

- Premium increases with higher risk
- Payouts are partially adjusted based on risk level
- Weekly payout caps ensure controlled financial exposure

6. Example Scenario (Vijayawada)

- Heavy rain triggers disruption
- Driver unable to work for 2 days

Medium Risk Plan:

- Daily payout = ₹325
- Total payout = ₹650
- Maximum payout capped at ₹700

7. System Workflow

1. User selects location
2. System fetches real-time data from APIs
3. Data is converted into risk scores
4. Disruption probability is calculated
5. Area is classified (Low / Medium / High risk)
6. Pricing model is applied
7. Payout is triggered automatically if conditions are met

Adversarial Defense & Anti-Spoofing Strategy

Market Crash Protection Protocol

To prevent fraudulent claims caused by GPS spoofing, VPN misuse, and coordinated bot attacks, Insura implements a **multi-layered verification system**. Instead of relying solely on device GPS, the system validates claims using **network data, device signals, behavioral patterns, and real activity data**. This ensures accurate payouts while maintaining a smooth experience for genuine users.

1. Multi-Layer Verification Approach

a. Network & IP Validation

- Compares **IP location with GPS location** to detect mismatches
- Flags claims originating from **VPNs, proxies, or data centers**
- Validates usage of **mobile carrier networks** (e.g., Jio, Airtel, Vi)

b. Behavioral & Movement Analysis

- Detects **impossible travel speeds** between delivery locations and claim areas
- Uses delivery platform data (e.g., Swiggy, Zomato) for verification
- Monitors **home zone consistency** to identify abnormal location jumps

c. Device & Sensor Verification

- Checks **GPS stability, altitude, and natural signal variations (jitter)**
- Uses **cell tower and Wi-Fi triangulation** for cross-verification
- Identifies emulator-based or fake GPS patterns

d. Fraud Ring Detection

- Identifies **clusters of identical or near-identical locations**
- Detects **simultaneous mass claims** from suspicious sources
- Flags coordinated activity targeting payout exploitation

2. Risk-Based Claim Processing

Each claim is assigned a **risk score** based on verification results:

- **Low Risk:** Automatically approved
- **Medium Risk:** Temporarily held for review
- **High Risk:** Flagged for detailed investigation

3. Honest User Protection

To ensure fairness:

- Users are **not instantly banned** on suspicion
- Flagged claims are moved to a **manual review process**
- Workers may submit **real-time proof (e.g., live photos)**
- Past behavior is considered to reduce false positives

4. Key Outcome

This approach:

- Prevents **GPS spoofing and large-scale fraud attacks**
- Protects the **financial stability of the system**
- Ensures **fast payouts for genuine workers**
- Maintains a **balance between security and usability**

Workflow

Insura Application Workflow
Phase 1: Onboarding & AI Risk Assessment
Enter Details
1 User Registration
Ravi Registers on the App.
2 AI Risk Calculation
AI Assesses Risk Level
→ Medium Risk Zone
Phase 2: Plan Selection & Policy Activation
MEDIUM RISK Plan
₹150/week
₹325/day
3 Plan Selection
Choose Insurance Plan
4 Payment & Activation
Policy Activated in MySQL
LPI
Phase 3: Active Monitoring
5 Continuous Monitoring
Real-Time API Polling
4 Disruption & Automated Claim Trigger
Heavy Rain Detected
Phase 4: Disruption & Automated Claim Trigger
6 Trigger Event
7 Fraud Validation
Location & Policy Check
Phase 5: Instant Payout & Resolution
8 Instant Payout
₹325 Sent to Ravi
EOP

Conclusion

This solution leverages AI and parametric insurance to provide a reliable income protection system for gig workers during unexpected disruptions. By combining predictive risk analysis, dynamic pricing, real-time event detection, and fraud prevention, the platform ensures fast, transparent, and fair payouts. Overall, it enhances financial security and resilience for gig workers while maintaining efficiency and scalability in the system.

Presentation video

https://youtu.be/mZ7x_GRFz0E